

Madiha Khan

Data Analyst

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SUMMARY

Detail-oriented Data Analyst with a strong foundation in SQL, Python, and Power BI. Skilled in data cleaning, data analysis, data visualization, and business intelligence, with hands-on experience in building dashboards and performing exploratory data analysis. Capable of transforming raw data into meaningful insights to support data-driven decision-making and improve business performance.

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Databases:** MySQL, SQL Server
- **BI Tools:** Power BI, DAX, Power Query
- **Data Visualization:** Dashboard Development, KPI Reporting
- **Techniques:** Data Cleaning, ETL, EDA, Data Modeling, Data Warehousing, Statistical Analysis, Data Validation, SQL Optimization
- **Tools:** Excel (Pivot Tables, VLOOKUP), SSMS, Power BI Desktop, Jupyter Notebook, Git

PROJECTS

CSV Intelligent Agent using Generative AI (Groq + LangChain)

- Developed a CSV AI Agent using LangChain Experimental (create_csv_agent) and ChatGroq (llama-3.3-70b-versatile) to enable natural language interaction with structured datasets
- Integrated Groq API using environment variables, ensuring secure and efficient LLM-based query processing
- Implemented dynamic query execution using agent.invoke(), enabling users to perform data analysis tasks such as filtering, aggregation, and insights generation
- Built custom query interfaces (ask() and smart_analyze() functions) to automate analysis workflows and generate AI-driven insights
- Utilized Pandas for CSV data handling and enabled intelligent data processing, improving analysis efficiency and automation

Text Intelligent Agent using Generative AI (Groq + LLM)

- Developed a Text AI Agent using LangChain Groq (ChatGroq) with llama-3.3-70b-versatile model for high-performance natural language processing
- Configured secure API integration using environment variables (GROQ_API_KEY), ensuring scalable and secure LLM access
- Implemented real-time query handling using .invoke() method, enabling dynamic and context-aware response generation
- Optimized model performance by setting temperature = 0, ensuring consistent and accurate outputs
- Built and tested the solution in Python (Jupyter Notebook), demonstrating automation, NLP capabilities, and API integration

Hospital Patient Visit & Treatment Analysis

- Analyzed 5,000+ patient records across 10+ attributes to identify healthcare trends and patient demographics
- Performed data validation ensuring 100% data completeness using SQL aggregations and CASE statements

- Evaluated treatment costs and patient stay duration, identifying average stay of 15 days
- Applied window functions (RANK) to identify top 5 high-cost treatments
- Generated insights on top diseases, hospital performance, and monthly trends across 12+ months

Customer Churn Analysis

- Analyzed 2,000+ customer records with 30+ features to identify churn patterns and risk segments
- Conducted exploratory data analysis (EDA) to uncover key business insights
- Developed ETL pipeline and handled missing data, improving data quality by 35%
- Built interactive dashboards for churn rate, revenue trends, and customer segmentation

Zapp Sales Analysis

- Analyzed 1,000+ sales records using Python (Pandas, NumPy) for business insights
- Performed data cleaning, preprocessing, and validation improving data quality by 25%
- Identified top-performing products contributing 40% of total revenue
- Delivered actionable insights to optimize sales strategy and decision-making

EXPERIENCE

Data Analyst Intern
Edversity LiveTechSkill, Nagpur, India

Dec 2025 – Present

- Performed data cleaning, preprocessing, and ETL processes using Python and SQL
- Conducted EDA to identify trends, patterns, and business insights
- Automated reporting and developed interactive Power BI dashboards, improving operational efficiency by 20%

Project Intern
Qaswa Technologies Pvt. Ltd., Nagpur, India

Dec 2024 – Feb 2025

- Designed and implemented database management system using MySQL
- Developed SQL queries, stored procedures, and reports to improve data accessibility
- Assisted in testing, debugging, and performance optimization

EDUCATION

B.Tech in Electronics & Telecommunication
R.T.M.N.U University, Nagpur, India

Nov 2022 – May 2025

Diploma in Computer Science
M.S.B.T.E, Nagpur, India

Aug 2019 – Aug 2022

AWARDS

- IEOM Undergraduate Student Paper Competition Award (Nov 2024)
- Recognized for Secure Database Access Management System

LANGUAGES

English, Hindi